

Final Exam

● Graded

Student

TIYA CHOKHANI

Total Points

96 / 100 pts

Question 1

PAGE 1

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Question 2

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Question 3

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Question 4

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Question 5

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Question 6

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Question 7

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Question 8

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Final ExamName: Tiya ChokhaniStudent ID: 305933966

By signing below, you affirm that you have neither given nor received unauthorized help on this exam.

Signature: Tiya**DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO!
YOU WILL GET A ZERO IF YOU DO.**

This exam contains 9 pages (including this cover page) and 12 problems. The exam will be graded out of 100 points. **This is an individual exam.** You are not allowed to use a calculator, any additional notes, look at anyone else's paper, have extra paper, talk to anyone, consult the internet, anything. It is considered cheating if you use additional notes, have wandering eyes, are caught talking, etc. Absolutely no headphones, smartwatches, or cellphones out! **I have a zero tolerance policy for cheating.**

Instructions: Print your name and student ID number above AND ON EVERY PAGE. You may use any available space on the exam for scratch work.

Good Luck!**To Receive Full Credit**

- Answer must be written in the box provided! Show your work on the rest of the page.
- You must show your work to receive credit.
- Proper Notation and Organized. Clear, concise, and correct use of mathematical notation is part of a correct answer. Work should be organized in a reasonable neat and coherent way in the space provided. Work scattered all over the page without a clearing ordering is not acceptable.

Do not write in the table to the right.

Page	Points	Score
1	15	13
2	15	4.5
3	10	10
4	15	15
5	10	10
6	10	10
7	15	13.5
8	10	10
Total:	100	

1. (5 points) Compute the eigenvalue(s) and eigenfunction(s) of $\begin{cases} X'' = -\lambda X & 0 \leq x \leq \pi \\ X(0) = 0 & X(\pi) = 0 \end{cases}$

$\lambda_n = \boxed{n^2 \quad n=0,1,2,\dots}$ $X_n(x) = \boxed{\sin(nx)}$

Handwritten notes: $n=0$ is trivial soln which isn't an eigenfcn. $n \neq 0$ should be eigenfcn.

$$x'' + \lambda x = 0$$

$$x(x) = a_1 \cos(\sqrt{\lambda}x) + b_1 \sin(\sqrt{\lambda}x)$$

$$x(0) = a_1 = 0$$

$$x(\pi) = b_1 \sin(\sqrt{\lambda}\pi) = 0$$

$$\sqrt{\lambda}\pi = n\pi \quad n=0,1,2,\dots$$

$$x_n(x) = b_n \sin(nx) \quad b_n \in \mathbb{R}$$

2. (5 points) Consider $f(t, y) = t + y$. For the IVP $y' = t + y$, $y(0) = 1$, does the Picard–Lindelöf theorem guarantee a unique global solution? Here global means part (a) or (1) of the Theorem. Justify briefly.

$f = t + y$ is continuous everywhere & $\frac{\partial f}{\partial y} = 1$ is also continuous & bounded, so acc to P-L (i) this IVP has a unique global soln in $D: [-c, c] \times \mathbb{R}$ where $(0, 1) \in D$ as f is continuous & Lipschitz in y on D .

3. (5 points) Let $f(x) = \begin{cases} -x, & -2 < x < 0 \\ \frac{1}{2}, & 0 < x < 2 \end{cases}$ be a periodic function. What value does the Fourier series converge to for $x = 2$?

$$\begin{aligned} f(x=2) &= \frac{1}{2} [f(2^-) + f(2^+)] \\ &= \frac{1}{2} [f(x^-) + f(-2^+)] \\ &= \frac{1}{2} \left[\frac{1}{2} + (-2) \right] \end{aligned}$$

Handwritten note: algebra

$$\boxed{\frac{1}{2}}$$

4. (10 points) Determine whether the Picard–Lindelöf theorem guarantees (global or local) existence and uniqueness for the IVP

$$y' = t + \sin(y) \quad y(0) = \frac{\pi}{2}.$$

Justify which part of the theorem applies and where the unique solution is guaranteed.

Let $D: [-c, c] \times \mathbb{R}$ where $(0, \frac{\pi}{2}) \in D$ for any $c > 0$
 f is continuous everywhere on D

$$\frac{\partial f}{\partial y} = \cos(y) \leq 1 \quad \frac{\partial f}{\partial y} \text{ is continuous \& bounded everywhere}$$

$\therefore f$ is Lipschitz in y on D $\& \exists$ a unique sol'n α in $[-c, c]$, $c > 0$
 acc to P-L (i)

5. (5 points) Compute the inverse Laplace Transform of $\frac{3p+7}{(p^2+2p+5)(p^2-9)}$. If the result includes a convolution, just set up the integral.

$$\int_0^x \left[3 \cosh(3t) - \frac{7}{3} \sinh(3t) \right] \left[\frac{1}{2} e^{-x-t} \sin(2(x-t)) \right] dt \quad -0.5$$

$$\frac{3p+7}{(p^2+2p+5)(p^2-9)} = \frac{3p+7}{(p^2-9)(p^2+4)} = \left[3 \frac{p}{p^2-9} - \frac{7}{3} \frac{3}{p^2-9} \right] \left[\frac{1}{2} \frac{2}{(p+1)^2+4} \right]$$

A B

$$\mathcal{L}^{-1}[A] = 3 \cosh(3x) - \frac{7}{3} \sinh(3x)$$

$$\mathcal{L}^{-1}[B] = \frac{1}{2} e^{-x} \sin(2x)$$

$$\mathcal{L}^{-1}[f] = F(p) g(p)$$

$$= \left[3 \cosh(3x) - \frac{7}{3} \sinh(3x) \right] * \left[\frac{1}{2} e^{-x} \sin(2x) \right]$$

$$\left[\frac{1}{2} e^{-x} \sin(2x) \right]$$

- $$\mathcal{L}[f(n)] = \frac{e^{-2p}}{p^2} - \frac{e^{-4p}}{p^2} + \frac{3e^{-4p}}{p}$$

$$-x + 2 + 5$$

$$-x + 7$$

$$-x + 4 + 3$$

$$-(x-4)+3$$

$$\mathcal{L}[f(x)] = \frac{e^{-2p}}{p^2} - \frac{e^{-4p}}{p^2} + \frac{3e^{-4p}}{p}$$

- $$\frac{15}{6} x^3 e^{-2x}$$

$$\frac{15}{(p+2)^4} = \frac{A}{p+2} + \frac{B}{(p+2)^2} + \frac{C}{(p+2)^3} + \frac{D}{(p+2)^4}$$

$$A(p+2)^3 + B(p+2)^2 + Cp + 2C + D = 15$$

$$A(p^3 + 6p^2 + 12p + 8) + Bp^2 + 4Bp + 4B + Cp + 2C + D = 15$$

$$\cancel{A} = 0 \quad \cancel{B} = 0 \quad \cancel{C} = 0$$

$$\begin{aligned} & \hookrightarrow (p+2)^{-1} p^2 + 4p + 4 \\ & \quad - \underline{(p+2)^{-2}} \\ & \hookrightarrow -(-2)(p+2)^{-3} \\ & \quad - 6(p+2)^{-4} \end{aligned}$$

$$2^{-1} \left[\frac{15}{(p+2)^4} \right] = \frac{15}{6} x e^{3x-2x}$$

$$\frac{15}{6} x^3 e^{-2x} = (-1)^3 \frac{d^4}{dx^4} \left(\frac{1}{p+2} \right)$$

$$= -d \frac{2}{d\rho^2 (P+2)^3}$$

$$= - \frac{d^2}{dp^2} \frac{(-1)}{(p+2)^2}$$

$$\frac{-6}{(p+2)^4} = \frac{-15}{6} \frac{6}{(p+2)^4}$$

8. (15 points) Find the stationary function of

$$J[y] = \int_0^1 ((y')^2 + y^2) dx,$$

with the boundary conditions $y(0) = 0$, $y(1) = 1$.

$$y(x) = \frac{1}{e - e^{-1}} [e^x - e^{-x}]$$

$$\frac{\partial F}{\partial y} = 2y$$

$$\frac{\partial F}{\partial y'} = 2y'$$

$$\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial y'} \right) = 2y''$$

$$2y - 2y'' = 0$$

$$y'' = y$$

$$y'' - y = 0$$

$$\text{let } y = e^{rx}$$

$$e^{rx}(r^2 - 1) = 0$$

$$r = \pm 1$$

$$y(x) = C_1 e^x + C_2 e^{-x}$$

$$y(0) = C_1 + C_2 = 0 \Rightarrow C_2 = -C_1$$

$$y(1) = C_1 e + C_2 e^{-1} = 1$$

$$= C_1 [e - e^{-1}] = 1$$

$$C_1 = \frac{1}{e - e^{-1}}$$

$$y(x) = \frac{e^x}{e - e^{-1}} - \frac{e^{-x}}{e - e^{-1}} = \frac{1}{e - e^{-1}} [e^x - e^{-x}]$$

9. (10 points) Solve the following using Laplace Transforms

$$y'' + 4y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 0.$$

$$y(t) = \frac{1}{5}e^{-t} - \frac{1}{5}\cos(2t) + \frac{1}{10}\sin(2t)$$

$$p^2 y - py(0) - y'(0) + 4y = \frac{1}{p+1}$$

$$y[p^2 + 4] = \frac{1}{p+1}$$

$$y = \frac{1}{(p+1)(p^2+4)} = \frac{A}{p+1} + \frac{Bp+C}{p^2+4}$$

$$Ap^2 + 4A + Bp^2 + Bp + Cp + C = 1$$

$$A+B=0 \quad B+C=0 \quad 4A+C=1$$

$$B=-A \quad -A+C=0 \quad -4A+C=0$$

$$B = -\frac{1}{5} \quad A = \frac{1}{5} \quad \underline{5C = 1}$$

$$C = \frac{1}{5}$$

$$y = \frac{1}{5} \left(\frac{1}{p+1} \right) + \frac{1}{5} \left(\frac{-p+1}{p^2+4} \right) = \frac{1}{5} \left(\frac{1}{p+1} \right) - \frac{1}{5} \frac{p}{p^2+4} + \frac{1}{10} \frac{2}{p^2+4}$$

$$\mathcal{L}^{-1} \left[\frac{1}{p+1} \right] = e^{-t}$$

$$\mathcal{L}^{-1} \left[\frac{p}{p^2+4} \right] = \cos(2t)$$

$$\mathcal{L}^{-1} \left[\frac{2}{p^2+4} \right] = \sin(2t)$$

$$y(t) = \frac{1}{5}e^{-t} - \frac{1}{5}\cos(2t) + \frac{1}{10}\sin(2t)$$

10. (10 points) Find the Fourier series for the periodic function

$$f(x) = x \quad \text{on } [-2, 2].$$

You only need to simplify trig values that are nice and easily obvious like 0 or $(-1)^n$.

$$a_0 = \boxed{0} \quad a_n = \boxed{0} \quad b_n = \boxed{\frac{4}{n\pi} (-1)^{n+1}}$$

Full Solution: $f(x) =$

$$\sum_{n=1}^{\infty} \frac{4}{n\pi} (-1)^{n+1} \sin\left(n\pi \frac{x}{2}\right)$$

$$a_0 = \frac{1}{2} \int_{-2}^2 x \, dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_{-2}^2 = \frac{1}{2} \left[\frac{4}{2} - \frac{4}{2} \right] = 0$$

$$f(-x) = -x = -f(x) \text{ so odd func so } a_0 \neq a_n = 0$$

$$a_n = \frac{1}{2} \int_{-2}^2 x \cos\left(\frac{n\pi x}{2}\right) dx = \frac{1}{2} \left[\frac{2}{n\pi} \int_{-2}^2 \sin\left(\frac{n\pi x}{2}\right) dx \right] = \frac{1}{2} \left[\left(\frac{2}{n\pi}\right) \cos(n\pi) - \cos(-n\pi) \right] = 0$$

checking

$$b_n = \frac{1}{2} \int_{-2}^2 x \sin\left(\frac{n\pi x}{2}\right) dx = \frac{1}{2} \left[\frac{-2x}{n\pi} \cos\left(\frac{n\pi x}{2}\right) \right]_{-2}^2 + \frac{2}{n\pi} \int_{-2}^2 \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{1}{2} \left[\frac{-2(2)}{n\pi} \cos(n\pi) - \frac{4}{n\pi} \cos(-n\pi) \right]$$

$$= \left[\frac{-2}{n\pi} \cos(n\pi) - \frac{2}{n\pi} \cos(n\pi) \right] = \frac{-4}{n\pi} \cos(n\pi)$$

$$= \frac{-4}{n\pi} (-1)^n = \frac{4}{n\pi} (-1)^{n+1}$$

11. (15 points) Solve

$$\begin{cases} u_{xx} = u_{tt} & 0 < x < 2, t > 0 \\ u_x(0, t) = 0 & u_x(2, t) = 0 \\ u(x, 0) = x(2-x) & u_t(x, 0) = 0 \end{cases}$$

You must show all steps when arriving at your answer (don't jump to form of the final solution). You do NOT need to solve for the coefficients, however you MUST set up their corresponding equation(s).

$$u_n(x, t) = \sum_{n=1}^{\infty} C_n \cos\left(\frac{n\pi x}{2}\right) \cos\left(\frac{n\pi t}{2}\right) + \frac{1}{2} C_0$$

$$C_n = \int_0^2 x(2-x) \cos\left(\frac{n\pi x}{2}\right) dx$$

$$\text{let } u(x, t) = X(x)T(t)$$

$$X''T = T''X$$

$$\frac{X''}{X} = \frac{T''}{T} = -\lambda$$

$$X'' + \lambda X = 0 \quad \lambda > 0 \text{ for non trivial sol'n}$$

$$X_n(x) = a_n \cos(\sqrt{\lambda} x) + b_n \sin(\sqrt{\lambda} x)$$

$$X'_n(x) = -a_n \sqrt{\lambda} \sin(\sqrt{\lambda} x) + b_n \sqrt{\lambda} \cos(\sqrt{\lambda} x)$$

$$X'(0) = b_n \sqrt{\lambda} = 0 \Rightarrow b_n = 0$$

$$X'(2) = -a_n \sqrt{\lambda} \sin(2\sqrt{\lambda}) = 0$$

$$2\sqrt{\lambda} = n\pi \quad n = 0, 1, 2, \dots$$

$$\lambda = \left(\frac{n\pi}{2}\right)^2$$

$$X_n(x) = a_n \cos\left(\frac{n\pi x}{2}\right)$$

$$T'' + \lambda T = 0$$

$$T(t) = c_1 \cos(\sqrt{\lambda} t) + c_2 \sin(\sqrt{\lambda} t)$$

$$T'(t) = -c_1 \sqrt{\lambda} \sin(\sqrt{\lambda} t) + c_2 \sqrt{\lambda} \cos(\sqrt{\lambda} t)$$

$$T'(0) = c_2 \sqrt{\lambda} = 0 \Rightarrow c_2 = 0$$

$$T_n(t) = C_n \cos\left(\frac{n\pi t}{2}\right)$$

-1.5

$$u_n(x, t) = \sum_{n=1}^{\infty} C_n \cos\left(\frac{n\pi x}{2}\right) \cos\left(\frac{n\pi t}{2}\right)$$

$$u_n(x, 0) = \sum_{n=1}^{\infty} C_n \cos\left(\frac{n\pi x}{2}\right) = x(2-x)$$

$$C_n = \frac{2}{2} \int_0^2 x(2-x) \cos\left(\frac{n\pi x}{2}\right) dx$$

12. (10 points) Solve Laplace's equation in polar coordinates

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0,$$

on the disk $0 \leq r \leq 3$, $0 \leq \theta < 2\pi$, with the boundary condition

$$u(2, \theta) = 7 + 5 \cos(2\theta) - 8 \sin(3\theta).$$

You MUST solve for the coefficients.

$$u(r, \theta) = 7 + \frac{5}{4}r^2 \cos(2\theta) - r^3 \sin(3\theta)$$

$$\text{let } u(r, \theta) = R(r)T(\theta)$$

$$R''T + \frac{1}{r}R'T + \frac{1}{r^2}T''R = 0$$

$$\frac{R''}{R} + \frac{1}{r}\frac{R'}{R} + \frac{1}{r^2}\frac{T''}{T} = 0$$

$$r^2\frac{R''}{R} + r\frac{R'}{R} + \frac{T''}{T} = 0$$

$$\frac{T''}{T} = -r^2\frac{R''}{R} - r\frac{R'}{R} = -\lambda$$

$$T'' + \lambda T = 0 \Rightarrow$$

$$T(-\pi) = T(\pi) \quad (1)$$

$$T'(-\pi) = T'(\pi) \quad (2)$$

$$T(\theta) = a \cos(\sqrt{\lambda}\theta) + b \sin(\sqrt{\lambda}\theta)$$

(1) & (2) only satisfied when $\lambda = n^2$

$$T(\theta) = a \cos(n\theta) + b \sin(n\theta)$$

$$-r^2R'' - rR' + R\lambda = 0$$

$$r^2R'' + rR' - \lambda R = 0$$

$$R(r) = \begin{cases} c_0 + d_0 \log r & n=0 \\ c_n r^n + d_n r^{-n} & n \neq 0 \end{cases}$$

As we have continuity at $r=0$ $d_n=0$

$$R(r) = c_n r^n$$

$$u(r, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} r^n [a_n \cos(n\theta) + b_n \sin(n\theta)]$$

$$u(2, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} 2^n a_n \cos(n\theta) + 2^n b_n \sin(n\theta)$$

$$= 7 + 5 \cos(2\theta) - 8 \sin(3\theta)$$

$$\frac{a_0}{2} = 7 \Rightarrow a_0 = 14$$

$$2^n a_n = 5 \text{ when } n=2 \Rightarrow a_2 = \frac{5}{4}$$

$$2^n b_n = -8 \text{ when } n=3 \Rightarrow b_3 = \frac{-8}{8} = -1$$

$$a_n = \begin{cases} \frac{5}{4} & n=2 \\ 0 & \text{otherwise} \end{cases}$$

$$b_n = \begin{cases} -1 & n=3 \\ 0 & \text{otherwise} \end{cases}$$



